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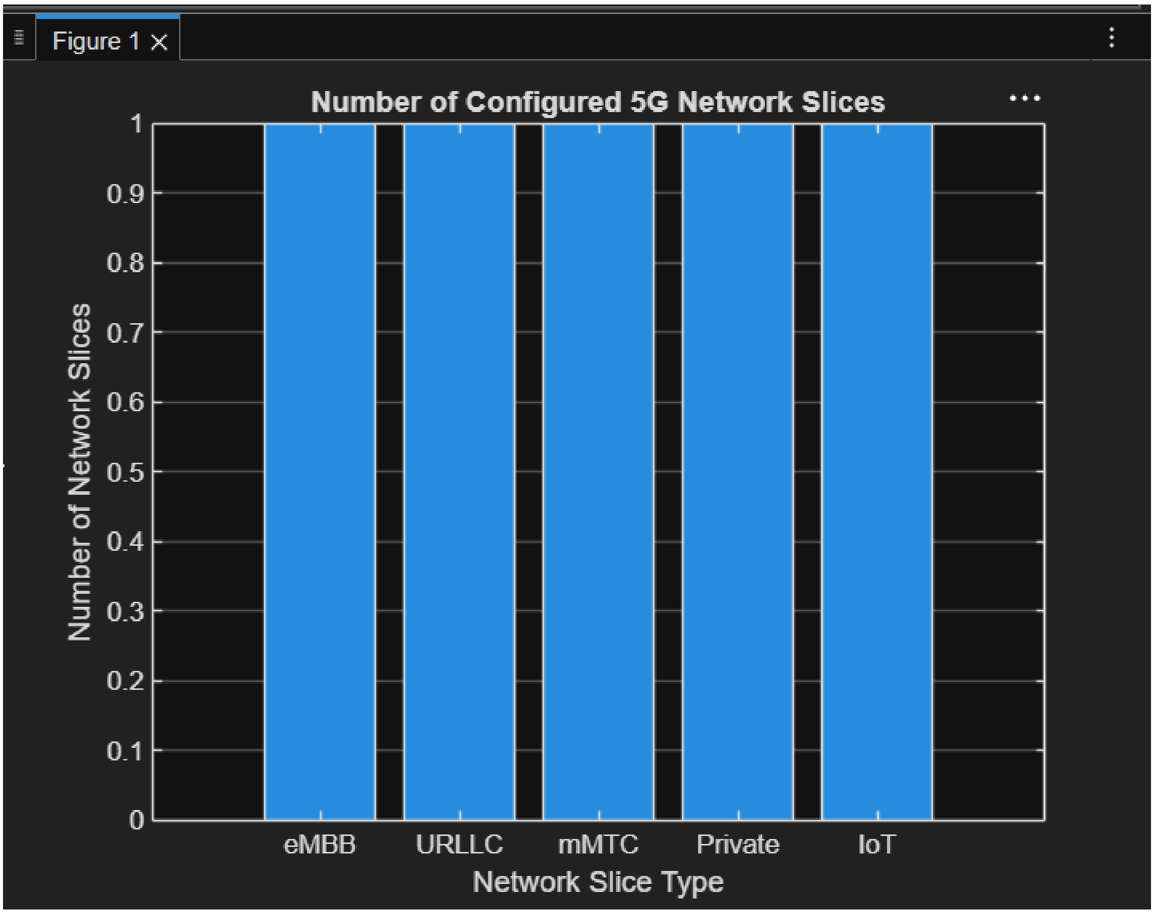
Experiment: 10

5G Network Slicing Environment Using MATLAB

OBJECTIVE 1 :

CODE:

```
clc;
clear;
close all;
slices = {'eMBB','URLLC','mMTC','Private','IoT'};
count = [1 1 1 1 1];    % One instance of each network slice
figure
bar(count)
grid on
set(gca,'XTickLabel',slices)
xlabel('Network Slice Type')
ylabel('Number of Network Slices')
title('Number of Configured 5G Network Slices')
```



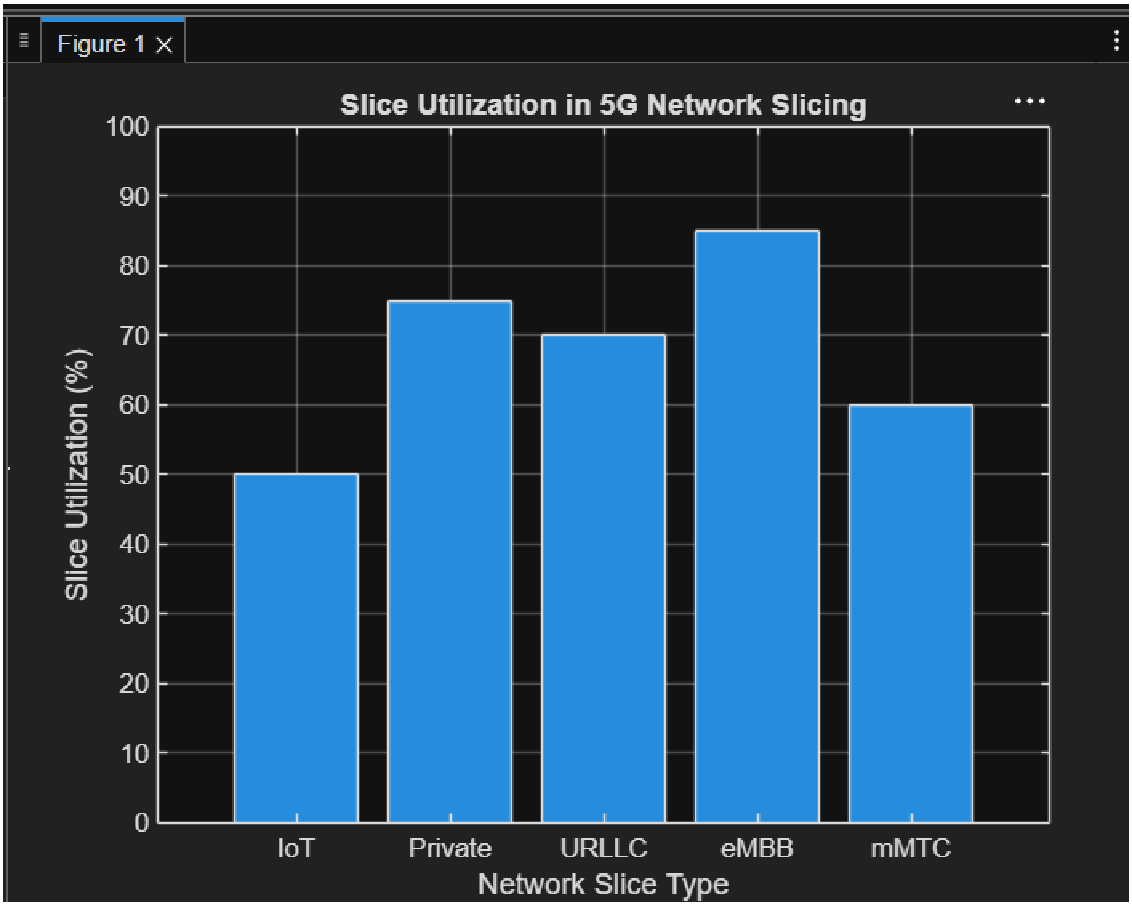
OBJECTIVE 2:

CODE:

```
clc;

clear;

close all;

slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});

utilization = [85 70 60 75 50];

figure;

bar(slices, utilization);

grid on;

xlabel('Network Slice Type');

ylabel('Slice Utilization (%)');
```

title('Slice Utilization in 5G Network Slicing');

ylim([0 100]);



OBJECTIVE 3:

CODE:

clc;

clear;

close all;

slices = [1 2 3 4 5];

latency = [25 5 40 15 60];

figure;

plot(slices, latency, '-o', 'LineWidth', 2, 'MarkerSize', 8);

```
grid on;

xticks(slices);

xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});

xlabel('Network Slice Type');

ylabel('End-to-End Latency (ms)');

title('5G Network Slicing: End-to-End Latency');

ylim([0 70]);
```